Trust Index Notes

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Abstract

Working notes on the definition of AI-judge trustworthiness, transcribed from handwritten pages.

1 Notation

Table 1: Notation (still provisional).

Symbol	Meaning
Items and agents	
x	candidate artifact
y^*	gold / reference artifact
$\hat{y}$, $\hat{y}(c)$	agent output on input c
A	set of agents
$a \in A$	an agent
B	set of behaviors
$b \in B$	a behavior
Judges / targets	
aj	AI judge configuration
θ_j, θ_{aj}	parameters of the AI judge (prompt, weights, …)
t_j	target (persona / rubric / human / …)
a_j	persona realized by configuration aj
h_j, H	human / target judge
π, Π	rater persona; set of personas
Scores and distributions	
$\hat{P}(s \mid \ldots, \theta_j)$	finite-sample score distribution of the AI judge
$\hat{P}^*(s \mid \ldots, \theta_* + t_j)$	finite-sample score distribution of the target
$\hat{P}_H$, $\hat{P}_{\mathrm{AI}}$	alternative ways of writing the two sides
Trust objects	
$\mathrm{TW}_m(aj)$	trustworthiness of aj (construct)
$\mathrm{TI}_m(aj)$	Trust Index (criterion; needs labels)
$\mathrm{PTI}_p(aj)$	Proxy Trust Index (instrument; label-free)

2 Definition from the first page

Definition 1 (Trustworthiness (handwritten form)).

$$\forall b \in B,\ \forall a \in A, \quad \text{if } t_j = a_j \quad \Rightarrow \quad \hat{P}\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_j\big) \;\approx\; \hat{P}^*\big(s \mid x,\, \hat{y}(c),\, y^*,\, \theta_* + t_j\big)$$

(the approximation measured by KL or total variation).

This says an AI judge is trustworthy if, for all behaviors, the judge matches the target judge (e.g. human, objective measure of that, specific person & strict, lenient, etc.).

Remark. The precondition $t_j = a_j$ is needed. If we set $t_j =$ strict but the concrete configuration a_j has parameters θ_{aj} (prompt, weights, human feedback, …) that map it onto a lenient judge, then the definition of TI_0 is not possible.

$\hat{P}^*$ denotes the target judge, but the star also carries the finite-sample nature of the estimate. We write $\hat{P}_H(\cdot)$ for the human target and $\hat{P}_\theta(\cdot)$ or $\hat{P}_{\mathrm{AI}}(\cdot)$ to distinguish the two sides.

In short: finite-sample AI judge versus finite-sample human target.

3 Notes from the second page

I feel I need to make clear that this is a "sensation game"—I am trying to make it clear with different words.

By the claim: the word Preference should mean "want" / goal, not some target / objective. Examples → human preferences.

Like how do I respond to the question "I prefer X but Y is the trustworthy answer"? I feel this isn't right.

My claim is that subjectivity doesn't exist once we agree on a frame of reference.

- (i) It can be modeled.
- Indeed, feelings can be modeled with probability theory.